

WILBERT ANDREW YONATHAN

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CAREER OBJECTIVE

Fourth-year Artificial Intelligence Engineering student seeking an AI Engineering or Computer Vision internship to apply deep learning, model optimisation and reinforcement learning to the development of efficient AI solutions. Available from 29 March 2027 for 3–5 months.

EDUCATION

Xiamen University Malaysia Sep 2023 – Sep 2027 (Expected)

Bachelor of Engineering in Artificial Intelligence | CGPA: 3.77/4.00

Relevant coursework: Computer Vision, Deep Learning, Advanced Machine Learning, Natural Language Processing, Deep Reinforcement Learning, Advanced Data Analysis, Software Engineering, Linear Algebra, Principles of AI.

PROJECT EXPERIENCE

DenseNet-121 Backbone Integration in YOLOv8n Apr 2026 – Present

for Paddy Disease Detection | Python, PyTorch, Ultralytics YOLO, OpenCV

- Developed a compact DenseNet-based backbone with SPPF and C2PSA for nine-class paddy disease detection, retaining the YOLOv8n neck and comparing against YOLOv8n and DenseNet-121 baselines.
- Prepared image annotations and dataset partitions, checked data quality, and conducted ablation studies using precision, recall, mAP@0.5, mAP@0.5:0.95, parameter counts and GFLOPs.
- Built and deployed an image-upload demo for three-model comparison, with aligned detection boxes, downloadable results, class-level charts and architecture efficiency comparisons.

GhostConv and SimSPPF Integration for Marine Organism Detection Jun – Jul 2026

Python, PyTorch, DU-MobileYOLO, OpenCV | URPC2020

- Integrated GhostConv into selected Multi-Concat convolutions and replaced the pooling module with SimSPPF, preserving the original deformable upsampling mechanism.
- Achieved 75.36% mAP@0.5 versus 74.23% for the retrained baseline, with 4.37M versus 4.70M parameters and 11.5 versus 12.1 GFLOPs; documented stricter-localisation trade-offs.
- Evaluated baseline and ablation variants and deployed a web demo with image uploads, bounding-box overlays, model comparisons and performance visualisations.

Autonomous dPickleball Agent Training May – Jul 2026

Python, PyTorch, Unity ML-Agents | Proximal Policy Optimisation (PPO)

- Trained and refined an autonomous PPO agent through curriculum learning, reward shaping and league self-play, experimenting with reward and penalty settings to improve gameplay behaviour.
- Contributed candidate policies and participated in round-robin evaluation and final agent selection, comparing checkpoints against agents with different playing strategies.
- Collaborated on the competition submission; the team placed second among 13 competing teams. Published the training workflow and linked the recorded competition.

PROJECT EXPERIENCE CONTINUED

ResNet Unified Convolution Network (Res_UCN) Nov – Dec 2025

- Contributed to a ResNet-18 adaptation using unified convolution modules and a custom Quinsh activation function to investigate efficient feature extraction and gradient propagation.
- Achieved 91.45% classification accuracy on CIFAR-10 with approximately 69% fewer parameters than standard ResNet-18; recorded training history and evaluated the accuracy-efficiency trade-off.

Facial Expression Recognition (FER2013) 2025

- Developed a custom CNN for seven-class facial expression recognition, achieving 66% accuracy and comparing against ResNet50, MobileNetV2 and VGG16 under consistent evaluation settings.
- Integrated Haar Cascade face detection into webcam inference and used confusion matrices, class-level metrics and Grad-CAM overlays to inspect predictions.

Alzheimer's Disease Prediction System 2024

- Developed a Streamlit application with clinical-data and MRI prediction pages, integrating trained models into an interactive inference and visualisation workflow.
- Compared classification models using accuracy, precision, recall and F1-score, and incorporated preprocessing and hyperparameter tuning into the experimental workflow.

EXTRACURRICULAR ACTIVITIES

Head of Photography and Videography | Love in Action Feb – Mar 2026

- Coordinated photo and video coverage, managed media workflows and edited outreach materials.

Head of General Affairs | XMUM AI Club Apr – Oct 2025

- Coordinated logistics and external collaborators for a national-level AI event.

General Affairs Member | AIESEC XMUM Sep 2023 – Sep 2025

- Supported committee administration and internal communication, coordinating tasks across committee activities.

ACHIEVEMENTS AND AWARDS

- Dean's List for three consecutive semesters; ranked within the top 25% of the class.
- XMUM Merit Scholarship awarded for the full four-year undergraduate programme, 2023–2027.

SKILLS AND LANGUAGES

Programming: Python, C, C++

AI and ML: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, CNN Design, Transfer Learning, Model Optimisation, Data Analysis

Frameworks and Libraries: PyTorch, torchvision, TensorFlow, Keras, scikit-learn, Ultralytics YOLO, Unity ML-Agents, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

Application Development: FastAPI, Gradio, Streamlit, Docker, Hugging Face Spaces, Render

Tools and Workflow: Git, GitHub, Jupyter Notebook, VS Code, Unity, Dataset Preparation, Ablation Studies, Model Evaluation

Languages: English, Mandarin, Indonesian, Malay, Hokkien

Soft Skills: Communication, Presentation, Teamwork, Leadership, Problem Solving

REFERENCES

Shaidah Binti Jusoh | Assistant Professor
Thesis Supervisor | Email: shaidahjusoh@gmail.com | Tel: [Phone]

Qasem Ashwaq Mukred Saeed | Computer Vision Supervisor
Email: [Email] | Tel: [Phone]